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| <b>BCO 086B</b> | <b>MACHINE LEARNING</b> | <b>3-0-1 [4]</b> |
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### Course Objectives

- To understand the basic concepts of learning and decision trees.
- To understand the neural networks and genetic algorithms
- To understand the Bayesian techniques
- To understand the instant based learning
- To understand the analytical learning and reinforced learning

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| <b>UNIT 1</b> | INTRODUCTION, CONCEPT LEARNING AND SUPERVISED LEARNING<br>ALGORITHMS: Introduction, Types of learning, Learning Problems – Designing Learning systems, Perspectives and Issues – Concept Learning – Version Spaces and Candidate Elimination Algorithm , Linear Regression Model, Naïve Bayes Classifier, Decision Tree, K Nearest Neighbor, Logistic Regression, Support Vector Machine, Random Forest Algorithm. |
| <b>UNIT 2</b> | UNSUPERVISED LEARNING ALGORITHM: Clustering- K-means Clustering, Hierarchical Clustering, Probabilistic Clustering, Apriori Algorithm, Association Rule Mining, Gaussian Mixture Model, Expectation Maximization. ENSEMBLE LEARNING-Bagging, Boosting and Stacking                                                                                                                                                 |
| <b>UNIT 3</b> | REGULARIZATION- Overfitting, Underfitting, Bias-Variance trade off, Cost Function, Regularized Linear Regression and Regularized Logistic Regression, Model Selection and train/Validation/Test Sets, VC Dimension. STATISTICAL LEARNING- Feature Extraction, Principal Component Analysis, Singular Value Decomposition, Feature Selection and subset selection.                                                  |
| <b>UNIT 4</b> | NEURAL NETWORKS AND GENETIC ALGORITHMS Neural Network Representation – Problems – Perceptron – Multilayer Networks and Back                                                                                                                                                                                                                                                                                        |

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|               | Propagation Algorithms – Gradient Descent.                                                                                                                                                                                                                                                                                                                                                      |
| <b>UNIT 5</b> | ANALYTICAL LEARNING AND REINFORCED LEARNING Perfect Domain<br>Theories – Explanation Based Learning – Inductive-Analytical Approaches - FOCL<br>Algorithm – Reinforcement Learning – Task – Q-Learning – Temporal Difference<br>Learning- Markov Decision Processes (MDP),<br>Introduction to Natural Language<br>Processing and Recommended System- Collaborative and Content based Filtering. |

### Course Outcome:

- CO1. Choose the learning techniques with this basic knowledge.
- CO2. Apply effectively neural networks and genetic algorithms for appropriate applications
- CO3. Apply Bayesian techniques and derive effectively learning rules.
- CO4. Choose and differentiate reinforcement and analytical learning techniques

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| <b>BCO 089 B</b> | <b>MACHINE LEARNING LAB</b> | <b>0-0-2 [2]</b> |
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### Course Outcomes

- CO1. To implement regression and various learning algorithms
- CO2. Implement various classifiers and analyze those classifiers base on confusion matrix
- CO3. pattern recognition and machine learning theories
- CO4. To design neural network and test network for different dataset
- CO5.** To apply the ML theories to different applications like face

### List of Experiments

- Lab 1. Implement the CANDIDATE – ELIMINATION algorithm. Show how it is used to learn from training examples.
- Lab 2. Write a program to implement Linear Regression and Logistic Regression
- Lab 3. Implement the ID3 algorithm for learning Boolean-valued functions for classifying the training examples by searching through the space of a Decision Tree.
- Lab 4. Design and implement Naïve Bayes Algorithm for learning and classifying TEXT DOCUMENTS.
- Lab 5. Implement K-Nearest Neighbor algorithm to classify the iris data set. Calculate the score also.
- Lab 6. Write a program to implement Support Vector Machine. Also discuss the confusion matrix

and score of model.

Lab 7. Apply EM algorithm to cluster a set of data and also apply K-Means algorithm on the same

data set to compare two algorithms.

Lab 8. Build an Artificial Neural Network by implementing Back-Propagation algorithm and test

the same using appropriate data set.

Lab 9. Implement the Non-Parametric Locally Weighted Regression Algorithm in order to fit data points. Select appropriate data set for your experiment and draw graph.

Lab 10. Build a Face detection system to recognize faces in a frame or image. You can use OpenCV for this task.

**AI & ML**